Artificial Intelligence and commercial applications – Introduction

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Course Goals

- Introduction of AI technologies and their potential applications to commercial systems.

Materials

- Main textbook
  - Artificial intelligence – A modern approach
- Reference book
  - Artificial intelligence illuminated
  - Data mining: concepts, models, methods and algorithms
  - Introduction to machine learning
- Example programs (mainly in Java)
  - Search, ANN, CART(decision tree), clustering, genetic algorithm, regression, GMM, HMM, KNN, SVM, minimax algorithm

Grading Rules of this course

- Attendance : 15%
- Homework: 25%
- Mid-term: 30%
- Final project/report: 30%

AI - Fantasy or Reality?

- A Space Odyssey (2001) : HAL (Heuristically programmed Algorithmic computer)
  - Play chess (search)
  - Read lips (image/speech recognition)
  - Engaged in conversation with humans
  - Go mad
- AI (Steven Spielberg): a robotic boy
  - Intelligent as a human being
  - Feel emotions

Definition of Artificial Intelligence

- A branch of computer science that is concerned with the automation of intelligent behavior. (Luger)
- But, a more fundamental problem…
  - What is "intelligence"?
  - Can intelligence be inferred from observable behavior, or does it require evidence of a particular internal mechanism?
  - Can *intelligence* be measured?
  - How is knowledge represented in the nerve tissue of a living being?
  - What is self-awareness?
- It is more concerned with expanding the boundaries of computer science than with defining those borders

What is Intelligence?

- *Intelligence* is difficult to define, but could be observed in the real world
- Intelligent behaviors of the humans
  - Solve problems, use tools, cooperate, communicate, recognize voices/objects, think logically, learn things, be adapted to environments, create art works, invent things…
- Intelligent behaviors of the other creatures
- Intelligent mechanisms of the world (non-creatures)

Intelligence of Creatures

- Apes can …
  - Use tool to get banana
  - Buy coke from vending machine
- Dolphins can…
  - Understand “language” & communicate with human
  - Hunt for sardines through cooperation
- Dogs can …
  - Recognize its master and his voice
  - Do a few jobs
  - Read
- Parrots can …
  - Do arithmetic

Intelligence of Creatures

- Pigs
  - talk (Korea)
- Ants can …
  - Find their way home
  - Communicate with one other
  - Manage their society
- Trees can …
  - Grow and do not collapse
  - Adapt itself to the environment (cactus)
- Migrant birds can …
  - Identify their direction

Intelligence in Non-Creatures

- Examples
  - Blood
  - Heartbeat
  - Virus
  - Rain
- Is *intelligence* relevant to thinking?
  - These examples show that intelligent behavior does not necessarily to come from *thinking rationally* with knowledge or information consciously …
  - e.g. the ghosts in Pac-Man game

AI History

- Aristotle 384-322 B.C. : Syllogism
  - All men are mortal. Socratis is a man.
    → Socratis is mortal.
- Gottfried Leibniz, late 17th century
  - Developing a formal mathematical language for reasoning
- George Boole, 1815-1864
  - Boolean algebra
- Gottlob Frege, 1879,1884
  - First-Order Predicate Calculus
- Charles Babbage, 1950s
  - Analytic engine – first computer

AI History

- *Artificial Intelligence* was first used by John McCarthy at a conference in Dartmouth College, in Hanover, New Hampshire.
  - John McCarthy invented LISP programming language in 1958
- Newell and Simon (1957)invented the idea of the General Problem Solver (GPS)
  - To solve almost any logical problem
  - Means-ends analysis

Defining Intelligence: Turing Test

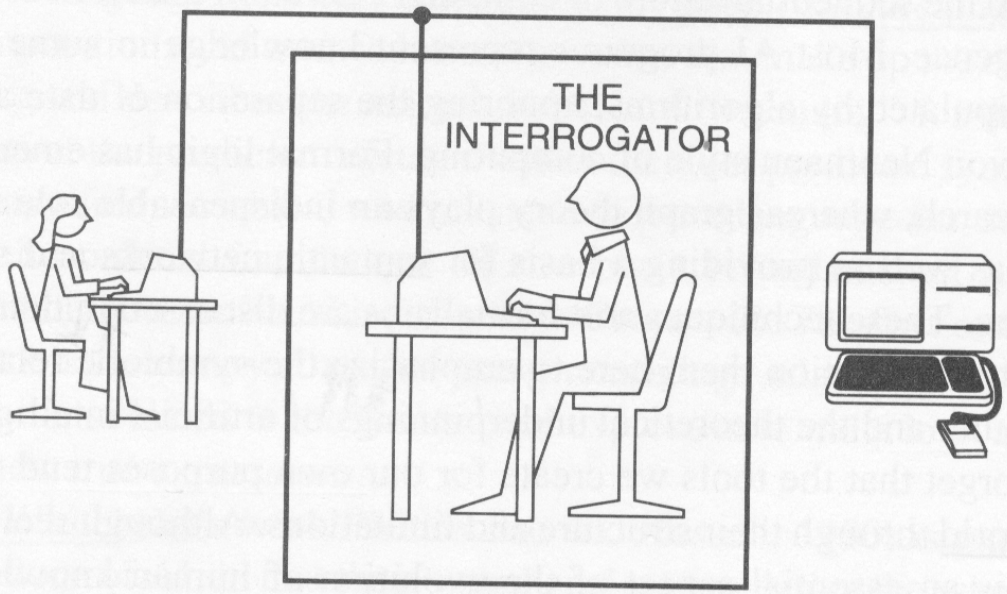


Figure 1.1 The Turing test.

- Can the interrogator tell the difference?
- Use human intelligence as a standard for the machine intelligence

Turing Test

- Alan Turing, *Computing Machinery & Intelligence*, 1950
- An Interrogator
  - Is given access to two individuals – a human and a computer
  - Can ask the two individuals any questions in text mode
  - Is requested to distinguish the computer from the human
- If the interrogator cannot distinguish the computer from the human, then the computer may be assumed to be intelligent.

Turing Test

- Giving an objective notion of intelligence
  - The behavior of a known intelligent being in response to a particular set of questions
- Preventing from being sidetracked by confusing & unanswered questions
  - whether the computer uses internal processes
  - whether the machine is conscious of its actions
- Eliminating bias in favor of living organisms over machine intelligence

Turing Test

- No program passed the Turing test yet
- Providing a basis for many schemes used to evaluate modern AI programs
- *Human behaviors* could be as indications of "intelligent" or "good"
- Giving an *operational* definition of "intelligence" (instead of descriptive definition)

Impacts of Turing Test

- Evaluation methodology for some researches
- Expert system
  - evaluated by human judgment
- Information retrieval
  - Reference answers are set by humans
- Speech synthesis
  - judgment for the quality of synthsized voices
- Spoken dialogue (wizard of Oz)
  - A human pretend to be a machine

Different Attitudes

- Strong AI
  - By giving a computer program sufficient processing power, and by providing it with enough intelligence, one can create a computer that can literally think and is conscious in the same way that a human is conscious.
  - A computer might have mind …
- Weak AI
  - *Intelligent behavior* can be modeled and used by computers to solve complex problems
  - it is not necessary to mimic human
    - Computers can be do quicker arithmetic operations
  - It is not necessary to imitating birds in order to fly
    - Just because a computer behaves intelligently does not prove that it is actually intelligent in the way that a human is.

Different Attitudes

- Weak AI
  - Artificial intelligence is the study of systems that act in a way that to any observer would appear to be intelligent.
  - Electronic dogs, fishes or dinosaur
- Strong AI
  - Artificial intelligence involves using methods based on the intelligence behavior of humans and other animals to solve complex problems.
  - Try to mimic humans or living organisms

Evolution of AI: 1960~1990

- Optimism → Realism
- Use "Divide and Conquer" strategies
  - Define some research issues for realization of machine intelligence
- No longer to create a robot as intelligent as a human (mission impossible?), but rather to use algorithms, heuristics, and methodologies based on the ways in which the human brains solves problems.

Milestones

- IBM Deep Blue, 11 May 1997
  defeat world champion of chess
- Robotcup soccer, since 1997
  http://www.robocup.org/
- IBM Watson: Question Answering Feb 2011
  - defeat human champion
  - natural language processing, information retrieval
- Apple Siri (Google voice), since 2011
  - Speech recognition, natural language processing
- Google automatic driver, 9 May 2012
  - sensor, classification, planning

Recent Achievement

- AlphaGo (deep mind)
  - Defeat mid-level amateur player
  - Try to challenge the smart professional player
  - Deep learning approach & game trees
- Go Game
  - Regarded to be a extremely difficult AI task
  - High complex and abstract
  - Similar to commercial behaviors
    - Benefits vs investments
    - Vague & uncertain
    - Live / dead
    - *Ko*

AI Researches

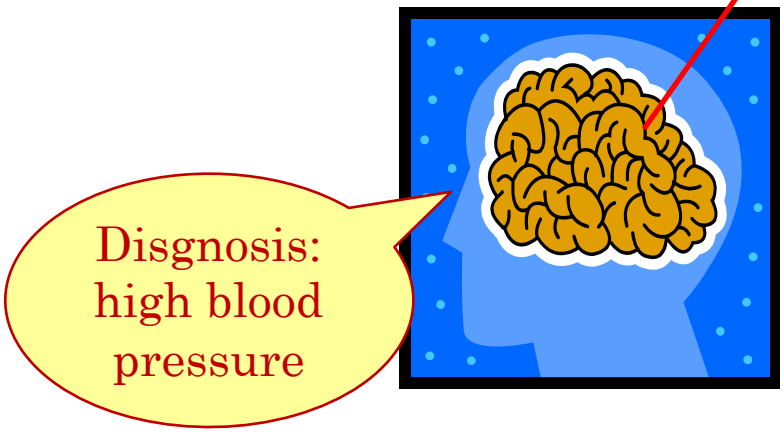
- Problem Solving
  - search , optimization, evolutionary computation
  - game playing, adversarial problem
- Reasoning
  - theorem proving, automatic reasoning, expert systems
  - fuzzy reasoning, probabilistic reasoning
  - knowledge representation, ontology, semantic network
- Learning
  - Induction, clustering, regression, summarization
  - Classification

Symbolic AI

- Make computers *think rationally*
  - Logic programming languages: LISP, PROLOG
  - Declarative instead of procedural
- Facts
  - made_from(cheese, milk).contains(milk, calcium).
- Rules
  - contains(X, Y):- made_from(X, Z), contains(Z, Y).
- After input with the facts and rules, the computer can answer such queries as
  - contains(cheese, calcium). → yes
    (does cheese contain calcium?)
  - contains(X, calcium) → X=milk. X=cheese.
    (what contains calcium?)

Reasoning

- Logic programming (Prolog/Lisp)
- Certainty factors
- Probabilistic reasoning
- Fuzzy logic



knowledge base

Rules:

if 舒張壓(DBP) 80~89 AND

收縮壓(SBP) 130~139

then 高血壓前期

if 舒張壓>90, 收縮壓>140

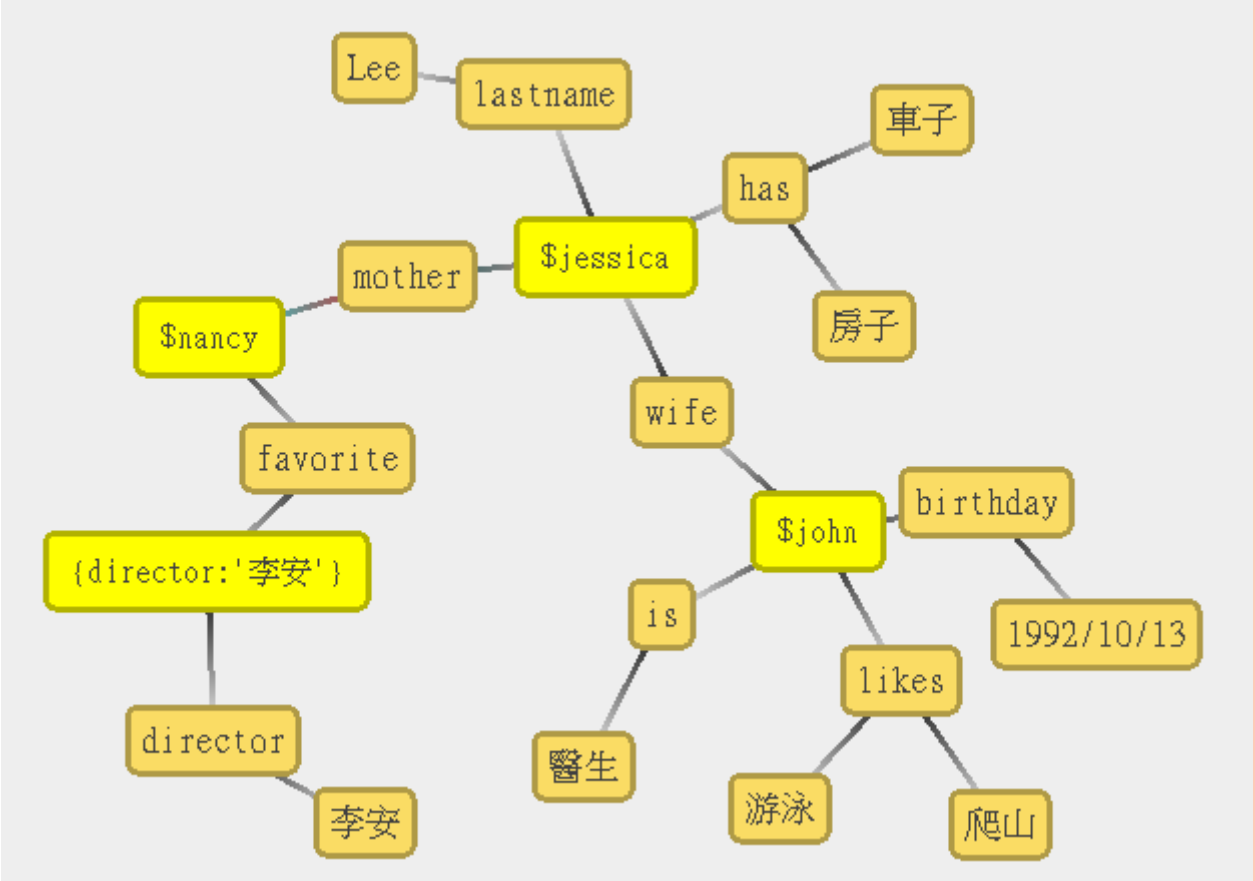
then 高血壓一期…

Facts:

舒張壓90,收縮壓148 → 高血壓

Knowledge Representation: Knowledge Graph

- Who is Jessica?
- Who among my friends likes to swim?
- What is the relationship between Jessica and John?



Relevant Researches

- Planning
  - robotics, scheduling
- Communication
  - natural language processing, information retrieval, dialogue systems
  - machine translation, multimedia retrieval
- Perception
  - speech recognition, machine vision
- Creation
  - evolutionary computation, automatic composition

Impacts of AI

- Explore the boundary of intelligence & inspire innovations
  - Definition of intelligence and extend the boundary
- Applicable tools and algorithms
  - Statistical learning、Search、Optimization、Evolutionary computation
- Joint disciplined research
  - Speech recognition/synthesis, machine vision, robot…
- Intelligent information systems
  - Ubiquitous

AI: Joint Disciplines

- Inspire many research directions
- Joint contributions from various areas
  - Philosophy, linguistics, cognitive psychology, neural science, biology, computer science, engineering, …

Impact From Philosophy

- Socratis and Plato
  - "I want to know what is characteristic of piety which makes all actions pious…"
  - An algorithm could be defined that described the behavior of humans determines whether a person's behavior was good or bad.
  - Is there more to the mind than simply a collection of neurons?
  - If each neuron in a human brain was replaced by an equivalent computation device, would the resultant be the same person?
- Rene Descartes (17th century)
  - Dualism: mind(soul) and matter
  - A cat or a dog is simply a complex machine

Impact From Linguistics

- Noam Chomsky 1950s
  - Syntactic Structure: a formal theory of the structure of human language
  - A structure of human knowledge

Impact From Psychology

- Cognitive psychology
  - Human brain uses knowledge or information that it is capable of processing in order to solve problems, make decisions, draw conclusions, and carry out other intelligent acts.
- Behaviorism
  - stimuli → behavior

Impact From Biology

- Neural Network
  - Learning capability
- Genetic Algorithm
  - Gene, chromosome
  - Evolution
- Speech Recognition
- Computer Vision

Different Points of View

Logic

Cognitive Science

Systems that think like humans	Systems that think rationally
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)
"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Systems that act like humans	**Systems that act rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998)
"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

Turing test
Knowledge rep.
NL processing
Machine learning
Automatic reasoning
robotics

Rational Agent

Issues covered in this course

- Problem solving
  - State space search
- Adversarial search
  - Game tree & min-max algorithm
- Local search
  - Genetic algorithm
  - Simulated annealing
- Classification
  - Decision tree
  - Artificial neural network
  - Bayesian network

Issues covered in this course

- Regression analysis
- Clustering
- Feature processing (PCA, LSA, RBM&AE)
- Gaussian mixture model (GMM) & Hidden Markov model (HMM)
- Reasoning
- Natural language processing
- Information retrieval and web Mining